

# Parameter-Flow, Gaussian, and Large-Deviation Frontiers for the $q$ -Fabius–Rvachev Family

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30 August 2026

## Abstract

Let  $U_0, U_1, \dots$  be independent  $\text{Unif}[-1, 1]$  variables and

$$X_q = (1 - q) \sum_{j \geq 0} q^j U_j, \quad 0 < q < 1.$$

At  $q = 1/2$  the density of  $X_q$  is the standard centered Rvachev up-function, and hence lies on the usual Fabius–Rvachev bridge. The current ProveIt corpus already contains an extensive theory of this family: its infinite-sinc transform, Bernoulli cumulants,  $q$ -Pochhammer factorizations, parameter limits, fixed-base endpoint asymptotics, finite prefix transport, and a first Gaussian correction. After a source-level crosswalk against that corpus, this report develops a different parameter layer.

First, for  $0 < r < s < 1$  we construct an explicit probability kernel  $p_{r,s}$  whose generating function is

$$P_{r,s}(z) = \frac{1-s}{1-r} \frac{1-rz}{1-sz}.$$

It transports the geometric coefficient profile at  $r$  to that at  $s$  and forms a convolution cocycle. A random tail shift with law  $p_{r,s}$  gives an explicit martingale coupling, proving the universal convex-order flow  $X_s \preceq_{\text{cx}} X_r$ . The kernels are the increments of an inhomogeneous compound-Poisson subordinator with geometric jump law.

Second, writing  $q = e^{-\delta}$ , we prove a full large-deviation principle at speed  $1/\delta$ . Its limiting scaled cumulant transform is

$$\Lambda(\theta) = \int_0^\infty \log\left(\frac{\sinh(\theta e^{-t})}{\theta e^{-t}}\right) dt = \int_0^\theta \frac{\log(\sinh u/u)}{u} du,$$

and the rate is  $I = \Lambda^*$ . The finite- $q$  transform is an exact midpoint Riemann sum. Midpoint Euler–Maclaurin therefore yields an all-orders expansion containing only even powers of  $\delta$  and Bernoulli polynomials at  $1/2$ ; in particular

$$\Lambda_\delta(\theta) = \Lambda(\theta) - \frac{\delta^2 \theta^2}{24} \Lambda''(\theta) + O(\delta^4).$$

The rate has an explicit rational Taylor series at the origin and a new Lambert- $W_{-1}$  boundary law as  $x \uparrow 1$ .

Third, we prove a uniform all-orders Edgeworth expansion for the *continuous* limit  $q \uparrow 1$ , valid after any fixed number of density derivatives. This is distinct from the discrete  $\text{Fup}_n$  Edgeworth theorem already present in the repository. Exact Bernoulli cumulants produce the Hermite coefficients, and a geometric infinite-sinc tail lemma supplies the global Fourier control. Numerical experiments reproduce the predicted orders 1, 2, 3, 4

and the midpoint orders 2, 4. The report closes with a moderate-deviation bridge, sharp saddlepoint and positivity conjectures, and a formalization roadmap. All theorem proofs here are paper-level arguments; no claim of Lean verification is made.

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# 1 Scope, corpus audit, and nonduplication boundary

## 1.1 What was compared

The source tree under

[https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/  
FabiusFunction/docs](https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs)

contains the expository Fabius and Rvachev documents, the Lean walkthrough, archival material, the canonical semi-formalized frontier volume, and a large set of current standalone drafts. The working comparison used the live source tree, the canonical frontier volume and its group READMEs, and the global draft manifest dated 28–30 August 2026 [4, 5]. The purpose of the comparison was not merely to avoid repeated wording: it was to remove mathematical layers whose main claims were already present under different titles.

The following table records the decisive exclusions and the boundary of the present contribution. “New here” always means *not found in the live repository crosswalk used for this report*; it is not an unqualified claim of priority over all mathematical literature.

| Corpus layer                                                    | Already present                                                                                                                                                                               | Layer developed here                                                                                                               |
|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Exponent- and $q$ -series frontiers, Parts VI–VII               | The geometric family, infinite sinc products, exact Bernoulli cumulants, $q$ -Pochhammer structure, quantitative $q \uparrow 1$ Gaussian convergence, and the first characteristic correction | A complete continuous-parameter Edgeworth hierarchy, uniform on $\mathbb{R}$ and after arbitrary fixed derivatives                 |
| Finite-sinc transport, Part V                                   | Convex order, peakedness, stop-loss identities, and exact metrics along the <i>prefix index</i> $N$                                                                                           | Convex order along the <i>geometric parameter</i> $q$ , with an explicit tail-shift martingale and a compound-Poisson co-cycle     |
| Atomic-function/Fup hierarchy                                   | A proved all-orders Edgeworth theorem for the discrete family $\text{Fup}_n$                                                                                                                  | A separate theorem for $X_q$ as the continuous parameter $q \uparrow 1$ ; the triangular arrays and cumulant weights differ        |
| Fixed-base endpoint/Lambert theory                              | Spatial endpoint asymptotics at fixed $q$ or fixed base, including Lambert- $W_{-1}$ inversion and periodic transform terms                                                                   | A parameter-boundary large-deviation rate as $q \uparrow 1$ , whose own $x \uparrow 1$ saddle has a distinct Lambert- $W_{-1}$ law |
| Shape, native orthogonal-polynomial, and noncommutative reports | Strict log-concavity, native Jacobi/Hankel theory, free and Boolean cumulants, and non-free-infinite-divisibility certificates                                                                | Explicitly omitted after the final cross-check                                                                                     |

## 1.2 Main claims of the report

The central claims are the following.

- R1.** An exact coefficient kernel  $p_{r,s}$ , its cocycle, its compound-Poisson generator, and an explicit martingale proof of  $X_s \preceq_{\text{cx}} X_r$ .
- R2.** An all-orders midpoint Euler–Maclaurin expansion for the scaled  $q \uparrow 1$  cumulant transform.
- R3.** A Gärtner–Ellis large-deviation principle with an explicit integral transform, rational central rate series, and Lambert- $W_{-1}$  edge inversion.
- R4.** A differentiated, uniform, all-orders Edgeworth expansion for the continuous geometric family.
- R5.** Matching between central and large deviations, together with saddlepoint, coefficient-positivity, and transseries conjectures.

The constructions extend verbatim to many bounded seed distributions. The uniform seed is singled out because it is exactly the Fabius–Rvachev case and because its cumulant transform is controlled by Bernoulli numbers.

## 2 The geometric-uniform law and the Fabius–Rvachev point

### 2.1 Random series, transforms, and fixed point

Let  $U_0, U_1, \dots$  be independent variables with density  $\frac{1}{2}\mathbf{1}_{[-1,1]}$ . For  $0 < q < 1$  define

$$X_q = (1 - q) \sum_{j=0}^{\infty} q^j U_j. \quad (1)$$

The series converges absolutely almost surely,  $\mathbb{E}X_q = 0$ , and  $|X_q| \leq 1$ . Its distribution is the unique compactly supported solution of

$$X_q \stackrel{d}{=} (1 - q)U + qX'_q, \quad (2)$$

where  $X'_q$  is an independent copy. We use

$$\text{sinc } z = \frac{\sin z}{z}, \quad \text{sinhc } z = \frac{\sinh z}{z},$$

with their continuous values at zero. The characteristic and moment generating functions are

$$\phi_q(t) = \mathbb{E}e^{itX_q} = \prod_{j=0}^{\infty} \text{sinc}((1 - q)q^j t), \quad (3)$$

$$M_q(t) = \mathbb{E}e^{tX_q} = \prod_{j=0}^{\infty} \text{sinhc}((1 - q)q^j t). \quad (4)$$

At  $q = 1/2$  the first product is  $\prod_{k \geq 1} \text{sinc}(t/2^k)$ , the standard Fourier transform of the centered Rvachev up-function. Thus  $\bar{X}_{1/2}$  is exactly the centered up-law; the usual piecewise shift/reflection relation then connects its density to the Fabius function [1, 2, 3].

## 2.2 Bernoulli cumulants

Let  $B_n$  denote the Bernoulli numbers with  $B_2 = 1/6$ . Since

$$\log \sinh z = \sum_{m=1}^{\infty} \frac{2^{2m-1} B_{2m}}{m(2m)!} z^{2m}, \quad |z| < \pi, \quad (5)$$

the even cumulants of  $U$  and  $X_q$  are

$$\kappa_{2m}(U) = \frac{2^{2m-1} B_{2m}}{m}, \quad (6)$$

$$\kappa_{2m}(X_q) = \frac{2^{2m-1} B_{2m}}{m} \frac{(1-q)^{2m}}{1-q^{2m}}, \quad \kappa_{2m+1}(X_q) = 0. \quad (7)$$

In particular

$$\sigma_q^2 := \text{Var}(X_q) = \frac{1-q}{3(1+q)}. \quad (8)$$

The moments are complete Bell polynomials in the cumulants [15], so every even moment is a rational function of  $q$  with Bernoulli input. This exact cumulant layer is part of the established corpus; our use of it below is to build the new parameter asymptotics.

## 2.3 The natural small parameter

Set

$$q = e^{-\delta}, \quad \delta = -\log q. \quad (9)$$

Then

$$1 - q = \delta s_\delta e^{-\delta/2}, \quad s_\delta := \frac{2 \sinh(\delta/2)}{\delta} = 1 + \frac{\delta^2}{24} + \frac{\delta^4}{1920} + O(\delta^6). \quad (10)$$

This exact half-step factorization is the source of the midpoint Euler–Maclaurin cancellation in [Section 4](#). Notice also

$$\sigma_q^2 = \frac{1}{3} \tanh \frac{\delta}{2} = \frac{\delta}{6} - \frac{\delta^3}{72} + \frac{\delta^5}{720} + O(\delta^7). \quad (11)$$

Thus the unstandardized law collapses to zero at scale  $\sqrt{\delta}$ , while its standardized form tends to a Gaussian.

## 3 An exact parameter-flow kernel and martingale transport

The finite-prefix transport theory already in the corpus orders successive truncations. Here the number of factors is infinite and the parameter itself moves. The key observation is that the geometric coefficient profile admits an exact stochastic factorization.

### 3.1 Coefficient cocycle

Write

$$a_j(q) = (1-q)q^j, \quad A_q(z) = \sum_{j \geq 0} a_j(q)z^j = \frac{1-q}{1-qz}. \quad (12)$$

**Theorem 3.1** (Parameter-flow kernel). *For  $0 < r < s < 1$ , define a probability mass function on  $\mathbb{N}_0$  by*

$$p_{r,s}(0) = \frac{1-s}{1-r}, \quad p_{r,s}(k) = \frac{(s-r)(1-s)}{1-r} s^{k-1}, \quad k \geq 1. \quad (13)$$

*Its probability generating function is*

$$P_{r,s}(z) = \sum_{k \geq 0} p_{r,s}(k) z^k = \frac{1-s}{1-r} \frac{1-rz}{1-sz}. \quad (14)$$

*The geometric coefficient profiles satisfy*

$$a(s) = p_{r,s} * a(r), \quad A_s(z) = P_{r,s}(z) A_r(z), \quad (15)$$

*and the kernels form the convolution cocycle*

$$p_{r,t} = p_{s,t} * p_{r,s}, \quad 0 < r < s < t < 1. \quad (16)$$

*Proof.* The coefficients in (13) are nonnegative and

$$p_{r,s}(0) + \sum_{k \geq 1} p_{r,s}(k) = \frac{1-s}{1-r} + \frac{s-r}{1-r} = 1.$$

Dividing the two rational functions in (12) gives

$$\frac{A_s(z)}{A_r(z)} = \frac{1-s}{1-r} \frac{1-rz}{1-sz},$$

whose geometric-series expansion is exactly (13). This proves (15). The cocycle follows immediately from  $P_{r,t} = A_t/A_r = (A_t/A_s)(A_s/A_r)$ .  $\square$

### 3.2 Compound-Poisson interpretation

The cocycle is not merely algebraic. Differentiating its logarithm in the upper parameter gives

$$\begin{aligned} \frac{\partial}{\partial s} \log P_{r,s}(z) &= \frac{z}{1-sz} - \frac{1}{1-s} \\ &= \sum_{k \geq 1} s^{k-1} (z^k - 1). \end{aligned} \quad (17)$$

Consequently

$$P_{r,s}(z) = \exp \left\{ \int_r^s \sum_{k \geq 1} u^{k-1} (z^k - 1) du \right\}. \quad (18)$$

**Proposition 3.2** (Inhomogeneous geometric-jump subordinator). *There exists an increasing process  $(N_q)_{0 < q < 1}$  with independent increments such that  $N_s - N_r$  has law  $p_{r,s}$ . At parameter value  $u$ , jumps occur with total intensity  $(1-u)^{-1} du$ , and conditional on a jump the size has geometric law*

$$\mathbb{P}\{J = k \mid u\} = (1-u)u^{k-1}, \quad k \geq 1. \quad (19)$$

*Proof.* Equation (18) is the Lévy–Khintchine formula for an inhomogeneous compound-Poisson process with jump intensity measure  $u^{k-1} du$  at size  $k$ . Summing over  $k$  gives total rate  $\sum_{k \geq 1} u^{k-1} = (1-u)^{-1}$ , and normalization gives (19).  $\square$

The expected number of jumps between  $r$  and  $s$  is  $\log((1-r)/(1-s))$ . Thus the apparently static operation of making the coefficient sequence more diffuse is governed by a simple pure-jump flow.

### 3.3 Explicit martingale coupling

The kernel acts by random shifts of the independent environment. This gives a direct convex-order proof, stronger than a comparison of finitely many moments.

**Theorem 3.3** (Tail-shift martingale transport). *Let  $V_0, V_1, \dots$  be iid integrable real random variables and define*

$$X_q^{(V)} = (1-q) \sum_{j \geq 0} q^j V_j. \quad (20)$$

*For  $0 < r < s < 1$ , let  $N$  be independent of the sequence and have law  $p_{r,s}$ . Put*

$$Y_{r,s} = \sum_{j \geq 0} a_j(r) V_{j+N}. \quad (21)$$

*Then*

$$Y_{r,s} \stackrel{d}{=} X_r^{(V)}, \quad \mathbb{E}[Y_{r,s} \mid V_0, V_1, \dots] = X_s^{(V)}. \quad (22)$$

*Consequently*

$$X_s^{(V)} \preceq_{\text{cx}} X_r^{(V)}. \quad (23)$$

*Proof.* For every fixed  $n$ , the tail  $(V_n, V_{n+1}, \dots)$  has the same joint law as the original iid sequence, hence the conditional-on- $N = n$  variable in (21) has law  $X_r^{(V)}$ . Thus  $Y_{r,s} \stackrel{d}{=} X_r^{(V)}$ . Conditioning instead on the entire environment and using Tonelli's theorem for the absolutely summable coefficients gives

$$\begin{aligned} \mathbb{E}[Y_{r,s} \mid (V_j)_{j \geq 0}] &= \sum_{n \geq 0} p_{r,s}(n) \sum_{j \geq 0} a_j(r) V_{j+n} \\ &= \sum_{k \geq 0} \left( \sum_{n=0}^k p_{r,s}(n) a_{k-n}(r) \right) V_k \\ &= \sum_{k \geq 0} a_k(s) V_k = X_s^{(V)}, \end{aligned}$$

where the penultimate equality is (15). Therefore  $X_s^{(V)}$  is the conditional expectation of a variable having law  $X_r^{(V)}$ . Jensen's inequality yields  $\mathbb{E}\Phi(X_s^{(V)}) \leq \mathbb{E}\Phi(X_r^{(V)})$  for every convex  $\Phi$  for which the expectations exist, which is exactly convex order.  $\square$

**Corollary 3.4** (Stop-loss and moment inequalities). *For  $0 < r < s < 1$  and every  $t \in \mathbb{R}$ ,*

$$\mathbb{E}(X_s - t)_+ \leq \mathbb{E}(X_r - t)_+. \quad (24)$$

*If the seed is centered and has finite variance, then*

$$\text{Var}(X_q^{(V)}) = \text{Var}(V) \frac{1-q}{1+q}, \quad (25)$$

*and this variance is strictly decreasing unless  $V$  is degenerate. All even convex moments that exist are nonincreasing in  $q$ .*



**Remark 3.5** (Why the proof is genuinely infinite-dimensional). The right shift of an infinite coefficient sequence is not a finite permutation. Nevertheless every shifted iid tail produces exactly the same weighted-sum law. Averaging over the shift kernel therefore supplies the martingale directly and bypasses finite-dimensional approximation. The finite-prefix convex-order chain in the repository and the parameter flow (23) are complementary rather than duplicate statements.

## 4 The scaled cumulant transform as a midpoint $q$ -integral

### 4.1 Exact half-step representation

For the uniform seed set

$$g(z) = \log \sinh z = \log \frac{\sinh z}{z}. \quad (26)$$

The scaled cumulant generating function at large-deviation speed  $1/\delta$  is

$$\Lambda_\delta(\theta) := \delta \log \mathbb{E} \exp\left(\frac{\theta X_{e^{-\delta}}}{\delta}\right). \quad (27)$$

Using (10), the  $j$ th argument of  $g$  is

$$\frac{\theta(1 - e^{-\delta})e^{-j\delta}}{\delta} = \theta s_\delta e^{-(j+1/2)\delta}.$$

Hence the exact identity

$$\Lambda_\delta(\theta) = \delta \sum_{j=0}^{\infty} g\left(\theta s_\delta e^{-(j+1/2)\delta}\right). \quad (28)$$

This is a midpoint Riemann sum on the logarithmic scale. It is the reason the first correction is  $O(\delta^2)$  rather than  $O(\delta)$ .

### 4.2 Limit transform

**Theorem 4.1** (Limiting scaled cumulant transform). *Uniformly for  $\theta$  in compact subsets of  $\mathbb{R}$ ,*

$$\Lambda_\delta(\theta) \longrightarrow \Lambda(\theta) := \int_0^\infty g(\theta e^{-t}) dt = \int_0^\theta \frac{g(u)}{u} du. \quad (29)$$

*The function  $\Lambda$  is even, real analytic, and strictly convex, with*

$$\Lambda'(\theta) = \frac{1}{\theta} \log \frac{\sinh \theta}{\theta}, \quad (30)$$

$$\Lambda''(\theta) = \frac{\theta \coth \theta - \log(\sinh \theta / \theta) - 1}{\theta^2} > 0, \quad (31)$$

$$\Lambda'(\mathbb{R}) = (-1, 1). \quad (32)$$

*At the origin,*

$$\Lambda(\theta) = \sum_{m \geq 1} \frac{2^{2m-2} B_{2m}}{m^2 (2m)!} \theta^{2m} = \frac{\theta^2}{12} - \frac{\theta^4}{720} + \frac{\theta^6}{17010} - \cdots. \quad (33)$$

*Proof.* For bounded  $\theta$ , the summand in (28) is smooth and bounded by  $C\theta^2 e^{-2t}$  after the identification  $t = (j + 1/2)\delta$ . Riemann-sum convergence is therefore uniform on compact

$\theta$ -sets and yields the first integral in (29). The change of variables  $u = \theta e^{-t}$  yields the second. Differentiation under the integral is justified by the same exponential envelope. Since  $g$  is the cumulant generating function of a nondegenerate uniform variable,  $g''(u) > 0$ ; therefore

$$\Lambda''(\theta) = \int_0^\infty e^{-2t} g''(\theta e^{-t}) dt > 0.$$

The explicit formulas follow by differentiating the second integral. Finally  $g(\theta) = \theta - \log(2\theta) + O(e^{-2\theta})$  as  $\theta \rightarrow +\infty$ , so  $\Lambda'(\theta) = g(\theta)/\theta \rightarrow 1$ ; oddness gives the lower endpoint. Integrating (5) termwise gives (33).  $\square$

### 4.3 All-orders midpoint Euler–Maclaurin expansion

Let  $B_n(x)$  be the Bernoulli polynomials and let  $D = \theta \frac{d}{d\theta}$  be the Euler operator.

**Theorem 4.2** (All-order midpoint expansion). *For every integer  $N \geq 1$  and every compact  $K \subset \mathbb{R}$ ,*

$$\Lambda_\delta(\theta) = \Lambda(\theta s_\delta) + \sum_{m=1}^{N-1} \frac{B_{2m}(1/2)}{(2m)!} \delta^{2m} D^{2m-1} g(\theta s_\delta) + O_{K,N}(\delta^{2N}) \quad (34)$$

*uniformly for  $\theta \in K$ . In particular the expansion contains only even powers of  $\delta$ . The first two nontrivial terms are*

$$\Lambda_\delta(\theta) = \Lambda(\theta) - \frac{\delta^2 \theta^2}{24} \Lambda''(\theta) + \delta^4 C_4(\theta) + O_K(\delta^6), \quad (35)$$

$$C_4(\theta) = \frac{7D^3 g - 10D^2 g + 5Dg - 2g}{5760}(\theta). \quad (36)$$

*Proof.* For  $f_{\delta,\theta}(t) = g(\theta s_\delta e^{-t})$ , all derivatives decay exponentially as  $t \rightarrow \infty$ , uniformly for  $\theta$  in a compact set. The midpoint Euler–Maclaurin formula [13] gives

$$\delta \sum_{j \geq 0} f_{\delta,\theta}((j + 1/2)\delta) = \int_0^\infty f_{\delta,\theta}(t) dt - \sum_{m=1}^{N-1} \frac{B_{2m}(1/2)}{(2m)!} \delta^{2m} f_{\delta,\theta}^{(2m-1)}(0) + O_{K,N}(\delta^{2N}). \quad (37)$$

The integral is  $\Lambda(\theta s_\delta)$ , while  $f^{(2m-1)}(0) = -D^{2m-1} g(\theta s_\delta)$ , giving (34). The vanishing  $B_{2m+1}(1/2) = 0$  explains the even-power structure.

For the first correction, use  $B_2(1/2) = -1/12$  and  $s_\delta = 1 + \delta^2/24 + O(\delta^4)$ :

$$\frac{\delta^2}{24} (g - Dg) = -\frac{\delta^2 \theta^2}{24} \Lambda''(\theta).$$

Expanding one order further and using  $B_4(1/2) = 7/240$  gives (36).  $\square$

**Remark 4.3** (A q-calculus reading). Equation (28) is a logarithmic Jackson-type sum whose mesh is  $\delta = -\log q$ , but the exact factor  $s_\delta$  centers the sampling at half steps. The resulting coefficients are Bernoulli polynomials at  $1/2$ , not merely Bernoulli numbers at zero. This provides a direct bridge between the repository’s  $q$ -product layer and its Bernoulli/Euler–Maclaurin layer [14].

## 5 A large-deviation principle as $q \uparrow 1$

### 5.1 The rate function

**Theorem 5.1** (Large-deviation principle). *The family  $X_{e^{-\delta}}$  satisfies a large-deviation principle on  $\mathbb{R}$  with speed  $1/\delta$  and good rate function*

$$I(x) = \sup_{\theta \in \mathbb{R}} \{\theta x - \Lambda(\theta)\}. \quad (38)$$

The function  $I$  is even, strictly convex on  $(-1, 1)$ , satisfies  $I(0) = 0$ , and is infinite for  $|x| \geq 1$ . For  $|x| < 1$ , let  $\theta_x$  be the unique solution of

$$x = \Lambda'(\theta_x) = \frac{1}{\theta_x} \log \frac{\sinh \theta_x}{\theta_x}. \quad (39)$$

Then

$$I(x) = \theta_x x - \Lambda(\theta_x), \quad I'(x) = \theta_x, \quad I''(x) = \frac{1}{\Lambda''(\theta_x)}. \quad (40)$$

*Proof.* The scaled logarithmic moment generating functions are exactly  $\Lambda_\delta$  in (27), and Theorem 4.1 gives pointwise convergence to the finite, differentiable function  $\Lambda$  on all of  $\mathbb{R}$ . Since every  $X_{e^{-\delta}}$  is supported on the common compact interval  $[-1, 1]$ , exponential tightness is automatic. The Gärtner–Ellis theorem [7, 8] therefore gives the LDP with Legendre transform (38). Strict convexity and (32) give the unique saddle and the parametric identities. Outside  $[-1, 1]$  the rate is infinite by support. At  $x = 1$ ,  $\theta - \Lambda(\theta) \rightarrow \infty$  by the asymptotic expansion proved in Theorem 6.1; hence  $I(1) = \infty$ , and similarly at  $-1$ .  $\square$

Thus for regular sets  $A$  the logarithmic probability scale is

$$\mathbb{P}\{X_q \in A\} = \exp \left[ -\frac{\inf_{x \in A} I(x) + o(1)}{-\log q} \right]. \quad (41)$$

This is a parameter-concentration law, not the fixed- $q$  spatial endpoint asymptotic already developed in the corpus.

### 5.2 Central rate series

The inverse saddle map in (39) is analytic at zero. Series reversion gives

$$\theta_x = 6x + \frac{36}{5}x^3 + \frac{1656}{175}x^5 + \frac{9504}{875}x^7 + \frac{709344}{67375}x^9 + \frac{18086976}{1990625}x^{11} + O(x^{13}), \quad (42)$$

$$I(x) = 3x^2 + \frac{9}{5}x^4 + \frac{276}{175}x^6 + \frac{1188}{875}x^8 + \frac{354672}{336875}x^{10} + \frac{1507248}{1990625}x^{12} + O(x^{14}). \quad (43)$$

All displayed coefficients are positive. The companion symbolic script checks positivity through the  $x^{16}$  coefficient.

**Conjecture 5.2** (Positivity of the rate series). *Every coefficient after the constant term in the Taylor expansion  $I(x) = \sum_{n \geq 1} c_n x^{2n}$  is strictly positive.*

This is stronger than convexity. A proof may require showing that the inverse of  $g(\theta)/\theta$  belongs to a positivity-preserving class, perhaps after a Stieltjes or Pick transform. The Bernoulli signs in (33) make coefficient positivity far from formal.

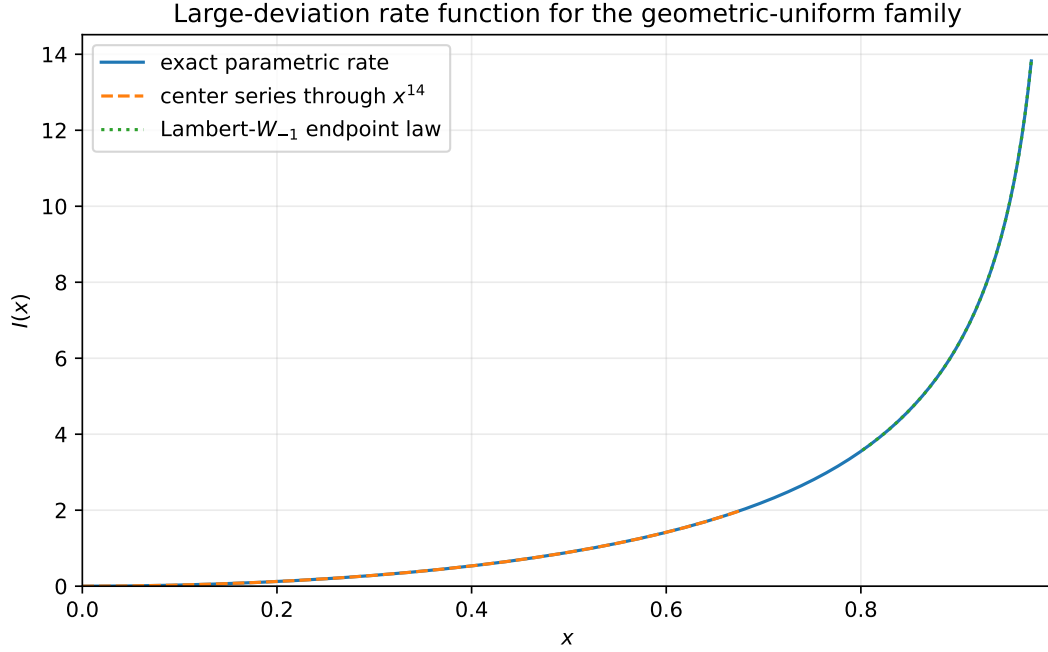


Figure 1: The exact parametric rate  $I(x)$ , its rational center expansion, and the Lambert- $W_{-1}$  boundary approximation. The latter is already visually indistinguishable from the exact curve for  $x \gtrsim 0.8$ .

## 6 Lambert- $W_{-1}$ behavior at the rate boundary

The fixed-base Fabius endpoint theory uses Lambert  $W$  to invert flat spatial tails. The rate function develops a second, conceptually different Lambert mechanism: the saddle parameter diverges as the deviation level approaches the fixed support boundary.

### 6.1 Asymptotics of the limiting transform

For  $u > 0$ ,

$$g(u) = u - \log(2u) + \log(1 - e^{-2u}). \quad (44)$$

Define the convergent constant

$$C_\Lambda = \int_0^1 \frac{\log(\sinh u/u)}{u} du - 1 + \int_1^\infty \frac{\log(1 - e^{-2u})}{u} du. \quad (45)$$

Numerically,

$$C_\Lambda = -0.96892042396303131827115715418362428628 \dots \quad (46)$$

**Theorem 6.1** (Transform and saddle endpoint). *As  $\theta \rightarrow +\infty$ ,*

$$\Lambda(\theta) = \theta - \frac{1}{2}(\log \theta)^2 - (\log 2) \log \theta + C_\Lambda + O\left(\frac{e^{-2\theta}}{\theta}\right), \quad (47)$$

$$1 - \Lambda'(\theta) = \frac{\log(2\theta)}{\theta} + O\left(\frac{e^{-2\theta}}{\theta}\right). \quad (48)$$

Let  $\eta = 1 - x \downarrow 0$  and

$$\theta_0(\eta) = -\frac{1}{\eta} W_{-1}(-\eta/2). \quad (49)$$

Then the exact saddle satisfies

$$\theta_{1-\eta} = \theta_0(\eta) \left( 1 + O\left( \frac{e^{-2\theta_0(\eta)}}{\log \theta_0(\eta)} \right) \right), \quad (50)$$

and, with  $L_\eta = \log \theta_0(\eta)$ ,

$$I(1-\eta) = \frac{1}{2} L_\eta^2 + (\log 2 - 1) L_\eta - \log 2 - C_\Lambda + o(1). \quad (51)$$

*Proof.* Integrate (44) from 1 to  $\theta$ . The elementary terms give  $\theta - \frac{1}{2}(\log \theta)^2 - (\log 2) \log \theta$ , while the two convergent remainders combine exactly into (45); the unintegrated tail is  $O(e^{-2\theta}/\theta)$ . Dividing (44) by  $\theta$  gives (48).

The saddle equation becomes

$$\eta\theta = \log(2\theta) - \log(1 - e^{-2\theta}). \quad (52)$$

Dropping the exponentially small last term yields  $\eta\theta = \log(2\theta)$ , equivalent to  $(-\eta\theta)e^{-\eta\theta} = -\eta/2$ . The diverging positive solution is (49) on branch  $W_{-1}$  [6]. A mean-value estimate in (52) gives (50). Finally substitute  $x = 1 - \eta$ , use  $\eta\theta = \log(2\theta) + o(1)$ , and subtract (47) from  $\theta x$ .  $\square$

**Remark 6.2** (Double logarithmic geometry). The standard branch expansion gives

$$-W_{-1}(-\eta/2) = \log(2/\eta) + \log \log(2/\eta) + o(1),$$

and consequently

$$L_\eta = \log(1/\eta) + \log \log(1/\eta) + o(\log \log(1/\eta)).$$

Thus the rate diverges primarily like  $\frac{1}{2} \log^2(1/\eta)$ , with a forced  $\log(1/\eta) \log \log(1/\eta)$  correction. This is the parameter-LDP counterpart of the log-squared endpoint structures elsewhere in the Fabius–Rvachev theory.

## 7 Uniform all-orders Edgeworth expansion for $q \uparrow 1$

### 7.1 Standardized infinite-sinc product

Set

$$Z_q = \frac{X_q}{\sigma_q}, \quad b_q = \frac{1-q}{\sigma_q} = \sqrt{3(1-q^2)}. \quad (53)$$

Then

$$\chi_q(t) := \mathbb{E} e^{itZ_q} = \prod_{j \geq 0} \text{sinc}(b_q q^j t). \quad (54)$$

Let  $\lambda_n(q)$  be the standardized cumulants of  $Z_q$ . From (7), for  $m \geq 2$ ,

$$\lambda_{2m}(q) = \frac{3^m 2^{2m-1} B_{2m}}{m} \frac{(1-q^2)^m}{1-q^{2m}} \quad (55)$$

$$= \frac{3^m 2^{3m-2} B_{2m}}{m} \frac{\sinh^m \delta}{\sinh(m\delta)}, \quad q = e^{-\delta}. \quad (56)$$

Therefore  $\lambda_{2m} = O(\delta^{m-1})$ . The first values are

$$\lambda_4 = -\frac{6}{5} \tanh \delta = -\frac{6}{5} \delta + \frac{2}{5} \delta^3 + O(\delta^5), \quad (57)$$

$$\lambda_6 = \frac{64}{7} \delta^2 - \frac{64}{7} \delta^4 + O(\delta^6), \quad (58)$$

$$\lambda_8 = -\frac{864}{5} \delta^3 + \frac{1728}{5} \delta^5 + O(\delta^7). \quad (59)$$

## 7.2 Weighted Hermite polynomial

Let  $H_n$  denote the probabilists' Hermite polynomials,  $\phi(x) = (2\pi)^{-1/2} e^{-x^2/2}$ , and let a multi-index  $\mathbf{k} = (k_2, k_3, \dots)$  have weight

$$\text{wt}(\mathbf{k}) = \sum_{m \geq 2} (m-1)k_m, \quad |\mathbf{k}|_* = \sum_{m \geq 2} m k_m. \quad (60)$$

For  $R \geq 0$  define

$$\mathcal{E}_{R,q}(x) = \sum_{\text{wt}(\mathbf{k}) \leq R} H_{2|\mathbf{k}|_*}(x) \prod_{m \geq 2} \frac{1}{k_m!} \left( \frac{\lambda_{2m}(q)}{(2m)!} \right)^{k_m}. \quad (61)$$

Only finitely many indices contribute. Keeping exact finite- $q$  cumulants is useful: it automatically resums many higher powers of  $\delta$ .

## 7.3 Global Fourier bounds

**Lemma 7.1** (Geometric infinite-sinc tails). *Fix  $q_0 \in (0, 1)$  and an integer  $L \geq 0$ . There exist positive constants  $c, C_L$  such that for  $q_0 \leq q < 1$ ,*

$$|\chi_q(t)| \leq \begin{cases} e^{-ct^2}, & |t| \leq b_q^{-1}, \\ C_L \exp\left(-\frac{c}{1-q}\right) (1 + b_q |t|)^{-L}, & |t| \geq b_q^{-1}. \end{cases} \quad (62)$$

*Proof.* Choose  $c_0 > 0$  such that  $|\text{sinc } u| \leq e^{-c_0 u^2}$  for  $|u| \leq 1$ . If  $|t| \leq b_q^{-1}$ , every factor lies in that range and

$$|\chi_q(t)| \leq \exp \left[ -c_0 b_q^2 t^2 \sum_{j \geq 0} q^{2j} \right] = e^{-3c_0 t^2}.$$

For  $|t| > b_q^{-1}$ , choose  $n$  so that  $b_q q^n |t| \leq 1 < b_q q^{n-1} |t|$ . The product of factors with  $j \geq n$  is at most

$$\exp \left[ -c_0 b_q^2 t^2 \frac{q^{2n}}{1-q^2} \right] \leq \exp \left[ -\frac{c_0 q_0^2}{1-q^2} \right],$$

which is bounded by  $e^{-c/(1-q)}$ . On the far part of the tail, the first  $L$  factors also satisfy

$|\operatorname{sinc} u| \leq |u|^{-1}$  and supply the stated polynomial decay. The bounded transition interval is absorbed into  $C_L$ .  $\square$

## 7.4 The differentiated theorem

**Theorem 7.2** (Continuous-parameter Edgeworth expansion). *Let  $f_q$  be the density of  $Z_q$ . For every pair of nonnegative integers  $R$  and  $r$ , there exist  $q_{R,r} < 1$  and  $C_{R,r} < \infty$  such that*

$$\sup_{x \in \mathbb{R}} \left| f_q^{(r)}(x) - \frac{d^r}{dx^r} [\phi(x) \mathcal{E}_{R,q}(x)] \right| \leq C_{R,r} \delta^{R+1}, \quad q_{R,r} < q < 1, \quad (63)$$

where  $\delta = -\log q$ .

*Proof.* Following the classical Fourier strategy for Edgeworth expansions [9, 10], we give the argument in enough detail to isolate the only genuinely nonstandard point: global control of the infinite sinc product.

Choose a small exponent  $\alpha > 0$ , depending on  $R$ , and put  $T_\delta = \delta^{-\alpha}$ , with  $\alpha < 1/4$ . On  $|t| \leq T_\delta$  the largest sinc argument is  $b_q|t| = O(\delta^{1/2-\alpha}) = o(1)$ . Expanding every logarithm by (5) with  $z$  replaced by  $iz$ , summing the geometric powers, and using (56) gives

$$\log \chi_q(t) = -\frac{t^2}{2} + \sum_{m=2}^{R+1} \frac{\lambda_{2m}(q)}{(2m)!} (it)^{2m} + \rho_{R,q}(t), \quad (64)$$

where, after decreasing  $\alpha$  if necessary,

$$|\rho_{R,q}(t)| \leq C_R \delta^{R+1} (1 + |t|^{2R+4}). \quad (65)$$

The same small-argument inequality used in Theorem 7.1 gives a uniform Gaussian envelope  $|\chi_q(t)| \leq e^{-ct^2}$  in this zone.

Exponentiating (64) and retaining all monomials of weight at most  $R$  produces exactly the Fourier polynomial corresponding to (61). Standard estimates for the exponential remainder therefore give

$$\left| \chi_q(t) - e^{-t^2/2} \mathcal{P}_{R,q}(it) \right| \leq C_R \delta^{R+1} (1 + |t|^{M_R}) e^{-ct^2}, \quad |t| \leq T_\delta, \quad (66)$$

for some finite  $M_R$ . Here  $\mathcal{P}_{R,q}$  is the same multi-index sum as (61), with  $H_n(x)$  replaced by the monomial  $(it)^n$ .

Fourier inversion after  $r$  derivatives is bounded in sup norm by the  $L^1$  norm of  $|t|^r$  times the Fourier error. The integral of (66) is  $O(\delta^{R+1})$ . On  $T_\delta < |t| \leq b_q^{-1}$ , the Gaussian part of Theorem 7.1 is  $O(\delta^N)$  for every  $N$ ; the Gaussian–polynomial approximant has the same property. On  $|t| > b_q^{-1}$ , take  $L > r + 1$  in Theorem 7.1; the contribution is  $O(e^{-c/(1-q)})$  times a power of  $b_q^{-1}$  and is again smaller than every power of  $\delta$ . This proves (63).  $\square$

## 7.5 First explicit corrections

Expanding the exact-cumulant form in the natural variable  $\delta$  gives

$$f_q(x) = \phi(x) \left[ 1 + \delta Q_1(x) + \delta^2 Q_2(x) + \delta^3 Q_3(x) + O_{C^r}(\delta^4) \right], \quad (67)$$

where

$$Q_1 = -\frac{1}{20}H_4, \quad (68)$$

$$Q_2 = \frac{4}{315}H_6 + \frac{1}{800}H_8, \quad (69)$$

$$Q_3 = \frac{1}{60}H_4 - \frac{3}{700}H_8 - \frac{1}{1575}H_{10} - \frac{1}{48000}H_{12}. \quad (70)$$

The absence of an  $H_4$  term in  $Q_2$  and of an  $H_6$  term in  $Q_3$  reflects the hyperbolic parity visible in (56). If one instead uses  $1 - q$  as the small parameter, these cancellations are obscured by the change of variables.

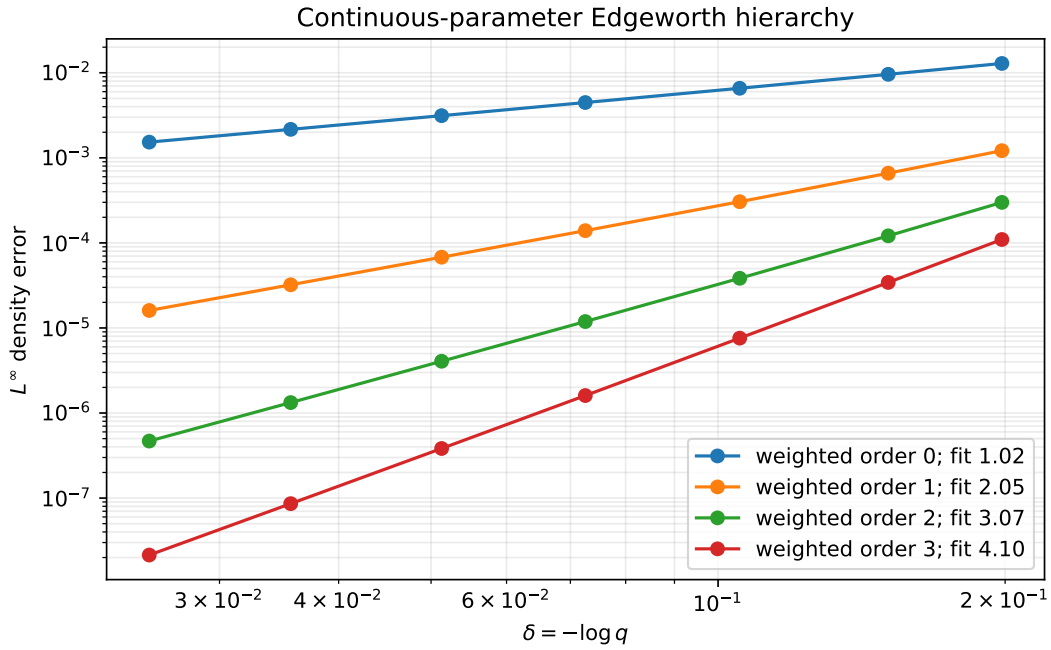


Figure 2: Sup-norm density errors from direct Fourier inversion. Fitted slopes 1.02, 2.05, 3.07, and 4.10 agree with Theorem 7.2.

## 8 Central, moderate, and large deviations in one scale diagram

The Edgeworth and large-deviation results are not separate asymptotic stories. The center expansion of  $I$  contains the same cumulant information as the Hermite correction.

Substitute  $x = \sigma_q z$  into (43) and use (11). Uniformly for bounded  $z$ ,

$$\frac{I(\sigma_q z)}{\delta} = \frac{z^2}{2} + \frac{\delta z^4}{20} + \delta^2 \left( \frac{23z^6}{3150} - \frac{z^2}{24} \right) + O(\delta^3(1 + |z|^8)). \quad (71)$$

The first exponential correction  $e^{-\delta z^4/20}$  matches the leading Hermite term  $-\delta H_4/20$  after the saddlepoint prefactor is expanded. The Gaussian moderate-deviation window suggested by the quartic term is

$$|z| = o(\delta^{-1/4}). \quad (72)$$

Beyond that scale, the rate function rather than a finite Edgeworth polynomial becomes the stable description.



**Principle 8.1** (Three nested regimes). As  $q = e^{-\delta} \uparrow 1$ :

- (i) fixed standardized  $z$  is governed by the differentiated Edgeworth series;
- (ii)  $1 \ll |z| \ll \delta^{-1/2}$  is governed by the Taylor expansion of  $I(\sigma_q z)/\delta$  and its moderate-deviation resummations;
- (iii) fixed  $x \in (-1, 1) \setminus \{0\}$  is governed by  $e^{-I(x)/\delta}$ , with Lambert- $W_{-1}$  behavior as  $x$  approaches the support edge.

**Conjecture 8.2** (Uniform sharp saddlepoint expansion). For every compact  $K \Subset (-1, 1)$ , the density of  $X_{e^{-\delta}}$  has an all-orders expansion

$$f_{X_q}(x) \sim \frac{e^{-I(x)/\delta}}{\sqrt{2\pi\delta\Lambda''(\theta_x)}} \left( 1 + \sum_{n \geq 1} c_n(x) \delta^n \right), \quad x \in K, \quad (73)$$

uniformly with all fixed  $x$ -derivatives. The coefficients should be rational differential polynomials in the tilted cumulants  $g^{(k)}(\theta_x e^{-t})$  integrated over  $t$ .

A proof should combine exponential tilting with a local Edgeworth theorem for the resulting infinitesimal triangular array. The midpoint expansion (35) implies that no order-one exponential amplitude is hidden in the finite- $q$  cumulant transform.

## 9 Numerical experiments and reproducibility

The archive contains two documented programs.

- `experiments.py` performs direct infinite-product Fourier inversion, midpoint cumulant-transform comparisons, rate-function and Lambert computations, and a Monte-Carlo stop-loss diagnostic.
- `symbolic_checks.py` uses exact SymPy rational arithmetic to verify cumulants, Hermite coefficients, the rate series, the kernel cocycle, and the first Euler–Maclaurin operators.

The main numerical outcomes are summarized in [Table 2](#).

Table 2: Observed convergence orders and consistency checks.

| Quantity                                          | Numerical result           |
|---------------------------------------------------|----------------------------|
| Uncorrected scaled-CGF order                      | 1.99999                    |
| After $-\delta^2 \theta^2 \Lambda''/24$           | 4.00134                    |
| Gaussian density error                            | 1.01783                    |
| Edgeworth weighted order 1                        | 2.05126                    |
| Edgeworth weighted order 2                        | 3.07181                    |
| Edgeworth weighted order 3                        | 4.09974                    |
| Endpoint constant $C_\Lambda$                     | −0.9689204239630311        |
| Minimum sampled stop-loss gap                     | 0 (to displayed precision) |
| Maximum sampled stop-loss gap, $r = .35, s = .72$ | 0.0757670                  |

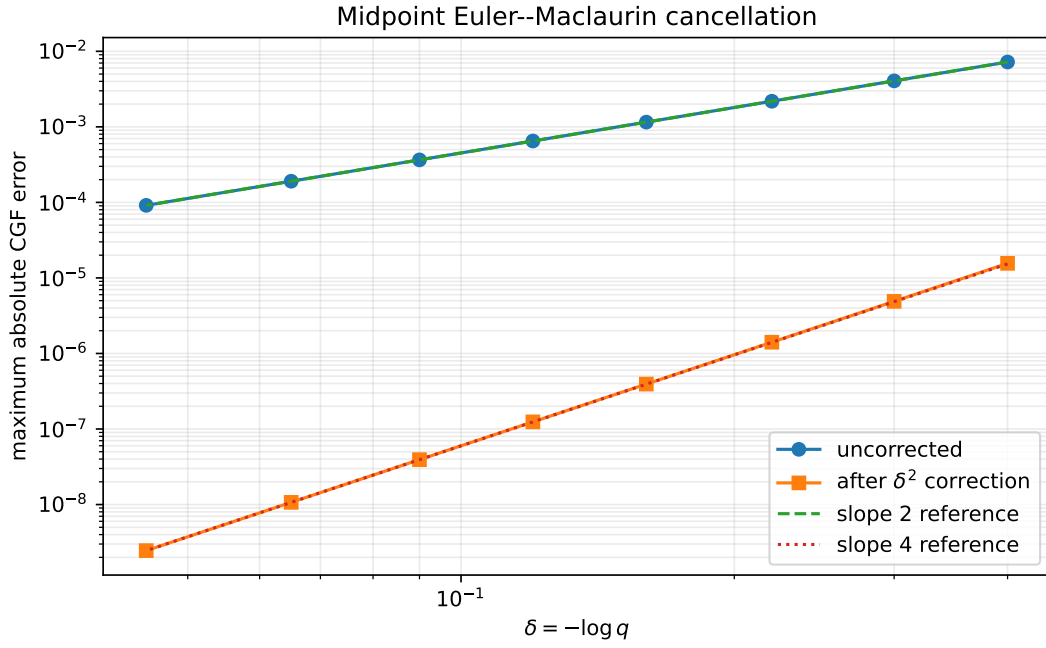


Figure 3: The raw midpoint-sum error is quadratic in  $\delta$ ; subtracting the proved first correction exposes the quartic remainder.

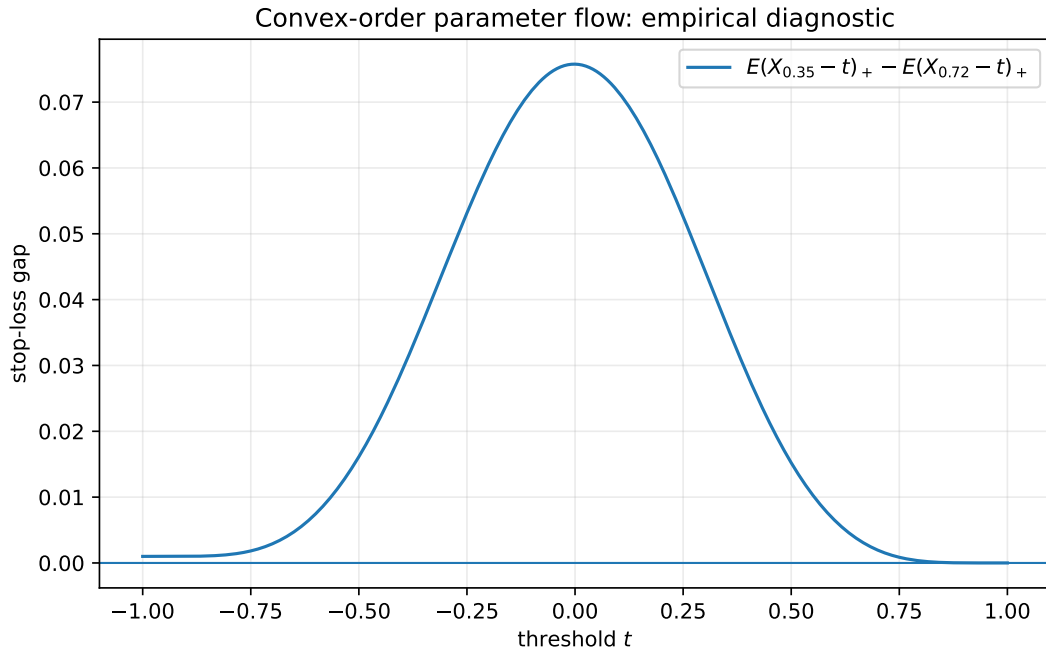


Figure 4: Empirical stop-loss gap for two parameter values. The plot is only a diagnostic; nonnegativity is proved exactly by the martingale coupling in [Theorem 3.3](#).

## 9.1 Numerical methodology

The standardized density is computed from

$$f_q(x) = \frac{1}{\pi} \int_0^\infty \chi_q(t) \cos(tx) dt.$$

The product is truncated only after the first omitted argument is below  $10^{-10}$ , and the quadratic logarithmic tail is inserted analytically. A cosine quadrature on  $[0, 30]$  then supplies the density. The values of  $|\chi_q(30)|$  stored in `data/edgeworth_errors.csv` are far below the reported approximation errors.

The finite- $q$  scaled cumulant transform is evaluated in its exact midpoint form (28). The endpoint constant is evaluated using the two absolutely convergent integrals in (45). The stop-loss experiment uses independent samples and a sorted-sample suffix-sum algorithm; it is not used in any proof.

## 10 Extensions, conjectures, and research program

### 10.1 General bounded seeds

The two structural results are not special to the uniform law.

**Proposition 10.1** (Seed-universal transform limit). *Let  $V$  be bounded and let  $K_V(\theta) = \log \mathbb{E} e^{\theta V}$ . For  $X_q^{(V)}$  from (20),*

$$\delta \log \mathbb{E} \exp \left( \frac{\theta X_q^{(V)}}{\delta} \right) \longrightarrow \Lambda_V(\theta) := \int_0^\infty K_V(\theta e^{-t}) dt. \quad (74)$$

*The same midpoint Euler–Maclaurin expansion holds with  $g$  replaced by  $K_V$ , and the family satisfies a Gärtner–Ellis LDP whenever the limiting transform is essentially smooth. Independently, the convex-order flow (23) holds for every integrable seed.*

*Proof.* The coefficient identity and martingale proof in Section 3 do not use the distribution of the seed. For the transform, the exact half-step calculation gives

$$\delta \sum_{j \geq 0} K_V(\theta s_\delta e^{-(j+1/2)\delta}),$$

and boundedness makes the Riemann-sum and Euler–Maclaurin arguments uniform. Gärtner–Ellis applies under its standard regularity hypotheses.  $\square$

This suggests a broad “geometric averaging universality class” indexed by the seed cumulant transform. The uniform seed is distinguished by the Bernoulli and infinite-sinc structures.

### 10.2 Sharp and beyond-all-orders questions

**Conjecture 10.2** (Interior all-orders saddlepoint theorem). *The expansion (73) is valid and Borel summable for each fixed compact subinterval of  $(-1, 1)$ . Its Borel singularities are governed by the complex zeros of  $\sinh z/z$ , hence by the same arithmetic zero lattice that controls the sinc-product Fourier theory.*

**Conjecture 10.3** (Endpoint transseries). *Let  $\eta = 1 - x$  and use the exact Lambert variable  $\theta_0 = -W_{-1}(-\eta/2)/\eta$ . The rate and saddle admit full transseries in inverse powers of  $\theta_0$  and exponentials  $e^{-2k\theta_0}$ , with no additional logarithmic scale after  $\theta_0$  is adopted. Matching this transseries to the finite- $q$  saddlepoint expansion should produce a uniform double-scaling theory when  $\eta$  and  $\delta$  tend to zero together.*

**Problem 10.4** (Critical boundary layer). *Determine the correct uniform asymptotics when*

$$1 - x \asymp \delta \log(1/\delta).$$

*This is the scale on which the Lambert saddle  $\theta_x \asymp 1/\delta$  meets the largest individual Fourier/MGF argument. A change of local model, and possibly a Stokes transition, should occur.*

### 10.3 Parameter-flow geometry

The exact shift subordinator invites several extensions.

**Problem 10.5** (Metric contraction along  $q$ ). *For which probability metrics  $d$  does the map  $q \mapsto \text{Law}(X_q)$  admit a closed differential law or sharp contraction coefficient? Convex order settles all stop-loss functionals, while the corpus settles several metrics along finite truncations; the parameter analogue for Wasserstein, Zolotarev, entropy, and Fisher metrics remains open.*

**Conjecture 10.6** (Strict stop-loss interior). *For the uniform seed and  $0 < r < s < 1$ ,*

$$\mathbb{E}(X_s - t)_+ < \mathbb{E}(X_r - t)_+$$

*for every  $t \in (-1, 1)$ . More generally, the difference should be strictly unimodal in  $t$  and inherit a total-positivity property from the random-shift kernel.*

**Problem 10.7** (Generator on laws). *Translate the coefficient jump generator*

$$\mathcal{L}_q F(n) = \sum_{k \geq 1} q^{k-1} [F(n+k) - F(n)]$$

*into a closed evolution equation for the density or its transform. The transform equation is explicit, but a useful physical-space equation may connect the parameter flow to the Stein–Koopman and diffusion layers already present in the corpus.*

### 10.4 Rate-series arithmetic

The rational coefficients in (43) arise from compositional inversion of a Bernoulli series. Their signs, denominators, and  $p$ -adic valuations appear to be a new arithmetic sequence.

**Problem 10.8** (Lagrange–Bernoulli formula). *Find a closed partition formula for the coefficient  $c_n$  in  $I(x) = \sum_{n \geq 1} c_n x^{2n}$ , and determine the exact denominator and 2-adic valuation. A useful formula should expose whether Theorem 5.2 follows from a hidden noncrossing, tree, or continued-fraction interpretation.*

### 10.5 Formalization roadmap

The parameter-flow theorem is especially suitable for formalization:

1. define the geometric coefficient sequence in  $\ell^1$  and prove the rational generating-function identity;
2. define  $p_{r,s}$  and prove positivity, normalization, and cocycle;
3. construct the random shift on an iid sequence and justify exchange of the two absolutely convergent sums;
4. apply conditional Jensen to obtain convex order;

5. formalize the midpoint identity as an algebraic rewrite before introducing analytic Euler–Maclaurin estimates.

The LDP and all-order Edgeworth proofs require substantially more analytic infrastructure. A practical formal route would first isolate the transform limit, strict convexity, and the first  $O(\delta^2)$  correction; the full uniform derivative theorem can then be modularized around [Theorem 7.1](#).

## 11 Conclusion

The geometric-uniform family is more than a deformation that happens to pass through the Fabius–Rvachev point. Its coefficient profile carries an exact probabilistic flow: increasing  $q$  is random tail shifting, and random tail shifting is martingale averaging. On the analytic side, the same parameter turns the logarithm of the  $q$ -product into a midpoint integral. That single observation explains the even Euler–Maclaurin hierarchy, produces a new large-deviation transform, and organizes the continuous Edgeworth expansion.

The resulting picture has three compatible scales. At the center, Bernoulli cumulants become Hermite corrections. At fixed deviations, their integrated limit becomes a Legendre rate. At the support edge, the saddle is inverted by  $W_{-1}$ . These layers are distinct from, but structurally aligned with, the finite-prefix transport, fixed-base endpoint, and discrete  $\text{Fup}_n$  results already in the repository.

## A A compact proof ledger

For ease of audit, the logical dependencies of the main results are:

| Result                      | Dependencies                                                                                  |
|-----------------------------|-----------------------------------------------------------------------------------------------|
| <a href="#">Theorem 3.1</a> | Rational generating functions only                                                            |
| <a href="#">Theorem 3.3</a> | <a href="#">Theorem 3.1</a> , iid tail invariance, conditional Jensen                         |
| <a href="#">Theorem 4.2</a> | Exact half-step factorization, midpoint Euler–Maclaurin, exponential derivative decay         |
| <a href="#">Theorem 5.1</a> | <a href="#">Theorem 4.1</a> , compact support, Gärtner–Ellis                                  |
| <a href="#">Theorem 6.1</a> | Elementary decomposition of $\log \sinh$ , $W_{-1}$ inversion                                 |
| <a href="#">Theorem 7.2</a> | Exact cumulants, analytic log-sinc expansion, <a href="#">Theorem 7.1</a> , Fourier inversion |

## B Explicit rate coefficients through order sixteen

The symbolic companion gives

$$\begin{aligned}
 I(x) = & 3x^2 + \frac{9}{5}x^4 + \frac{276}{175}x^6 + \frac{1188}{875}x^8 + \frac{354672}{336875}x^{10} \\
 & + \frac{1507248}{1990625}x^{12} + \frac{496585728}{766390625}x^{14} + \frac{53214713856}{65143203125}x^{16} + O(x^{18}),
 \end{aligned}$$

and

$$\begin{aligned}\theta_x = & 6x + \frac{36}{5}x^3 + \frac{1656}{175}x^5 + \frac{9504}{875}x^7 + \frac{709344}{67375}x^9 \\ & + \frac{18086976}{1990625}x^{11} + \frac{993171456}{109484375}x^{13} + \frac{851435421696}{65143203125}x^{15} + O(x^{17}).\end{aligned}$$

## C Files in the reproducibility package

The ZIP archive contains:

- `q_fabius_parameter_frontiers.tex` and the compiled PDF;
- `experiments.py`, `symbolic_checks.py`, and `requirements.txt`;
- CSV tables for the cumulant-transform, rate, Edgeworth, and stop-loss experiments;
- `numerical_summary.txt` and `symbolic_checks.txt`;
- the four figures in PDF and PNG formats;
- a short `README.md` with reproduction commands.

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